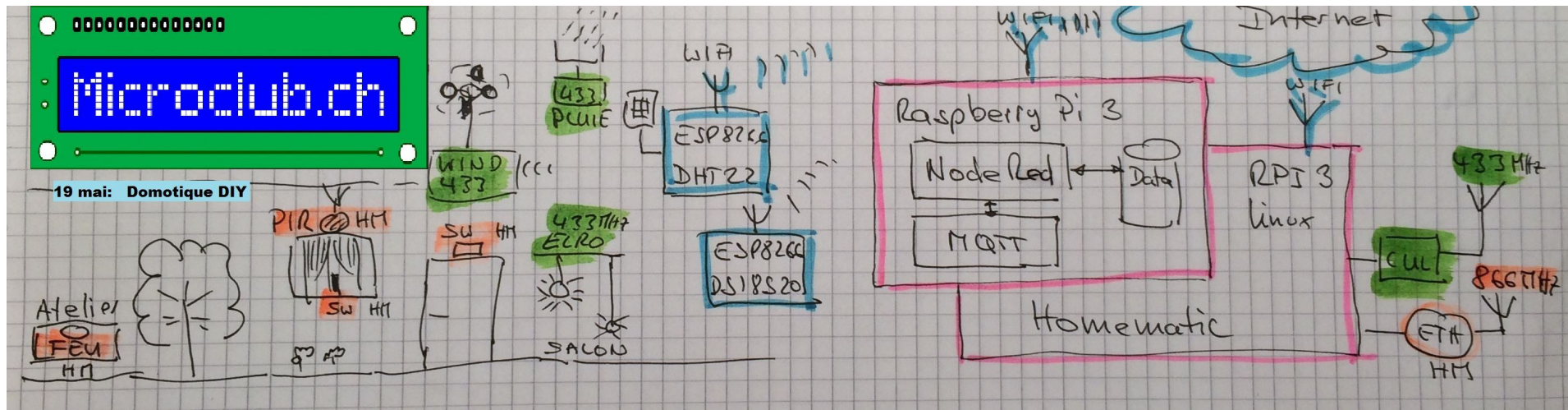


Domotique DIY

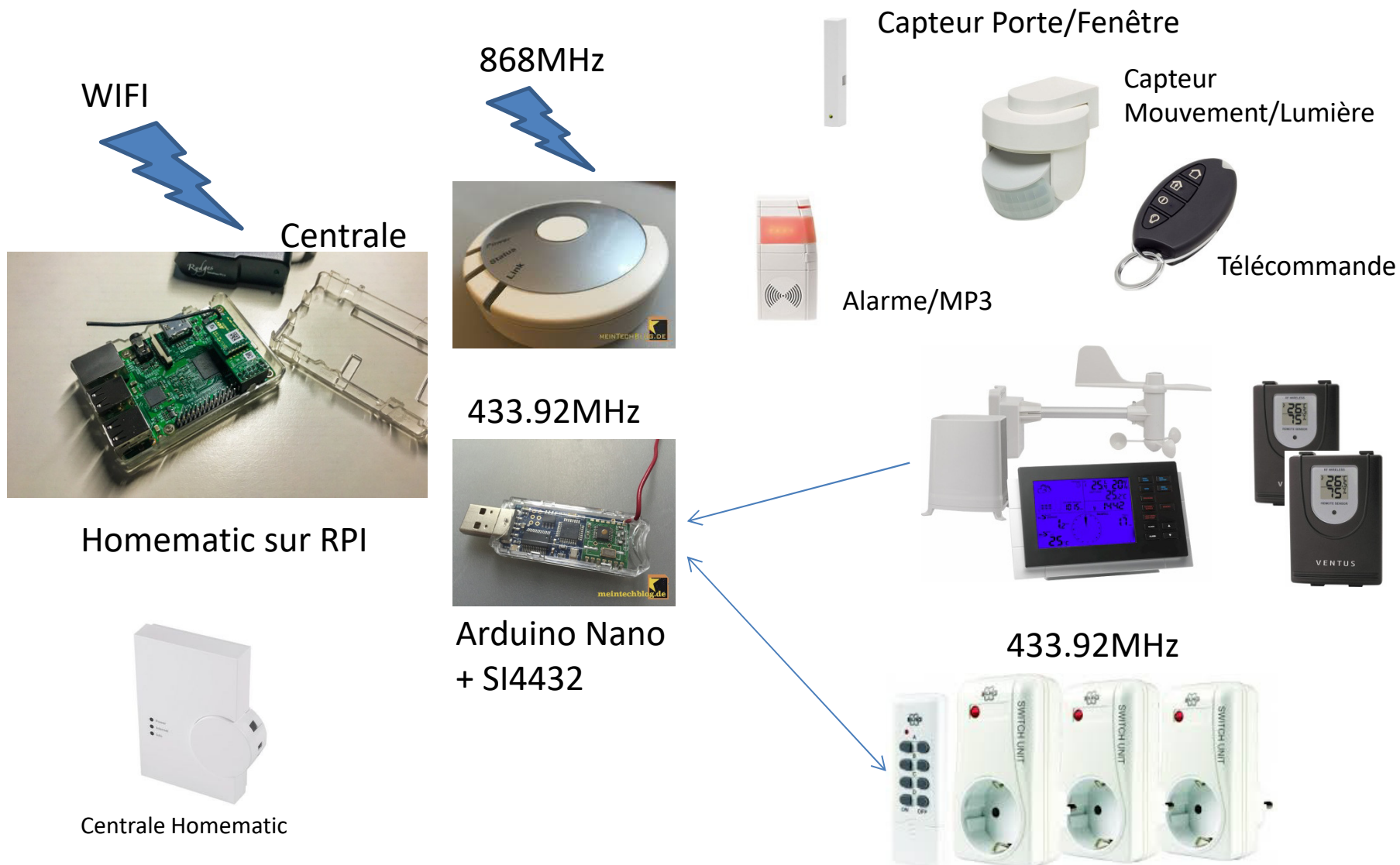
Mise à Jour 2018



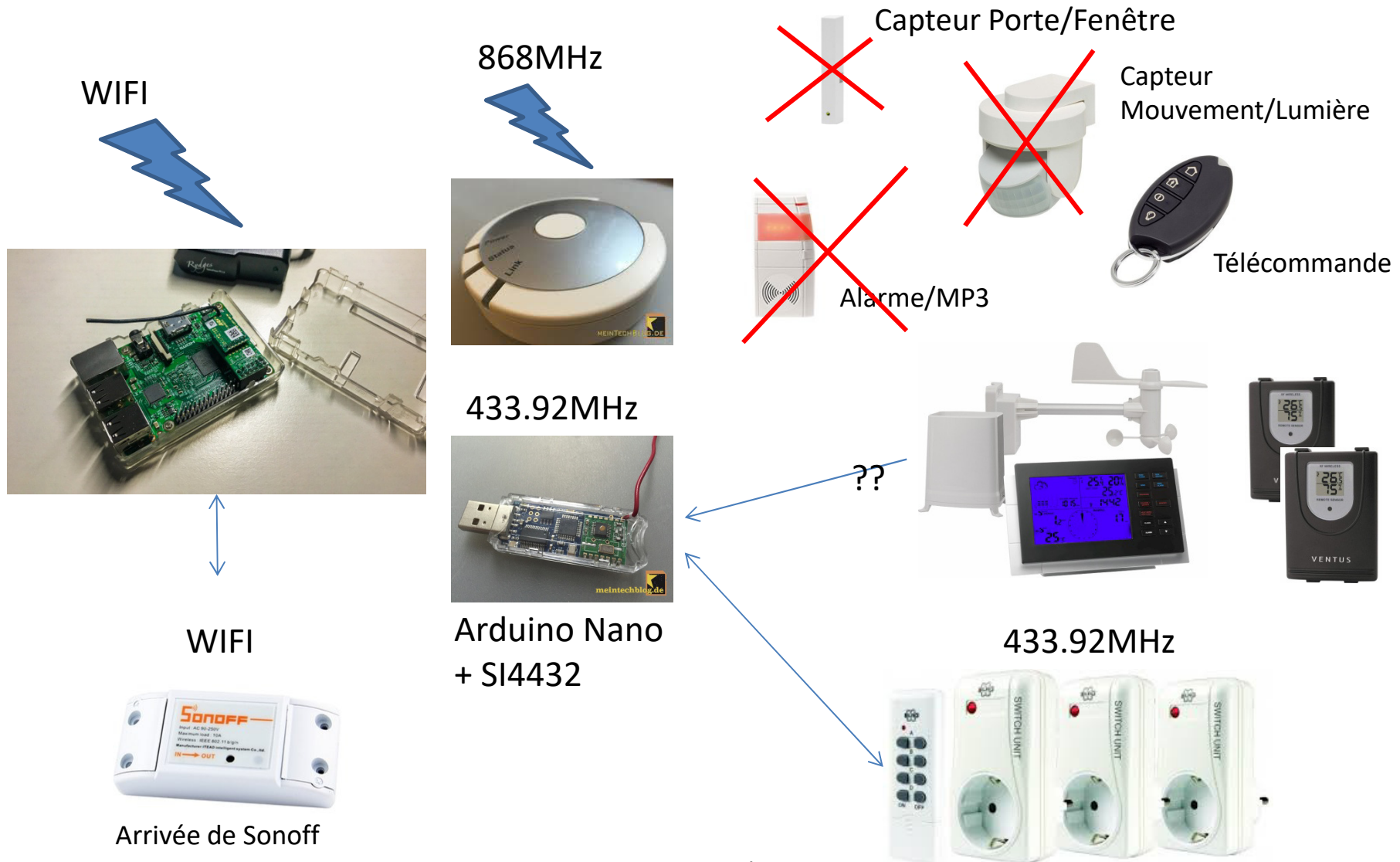
Programme de la soirée

- Introduction
 - Xiaomi, capteurs domotique low-cost
 - Intégration de Xiaomi dans Node-Red
-
- A suivre:
 - Construction d'un client MQTT pour notre domotique

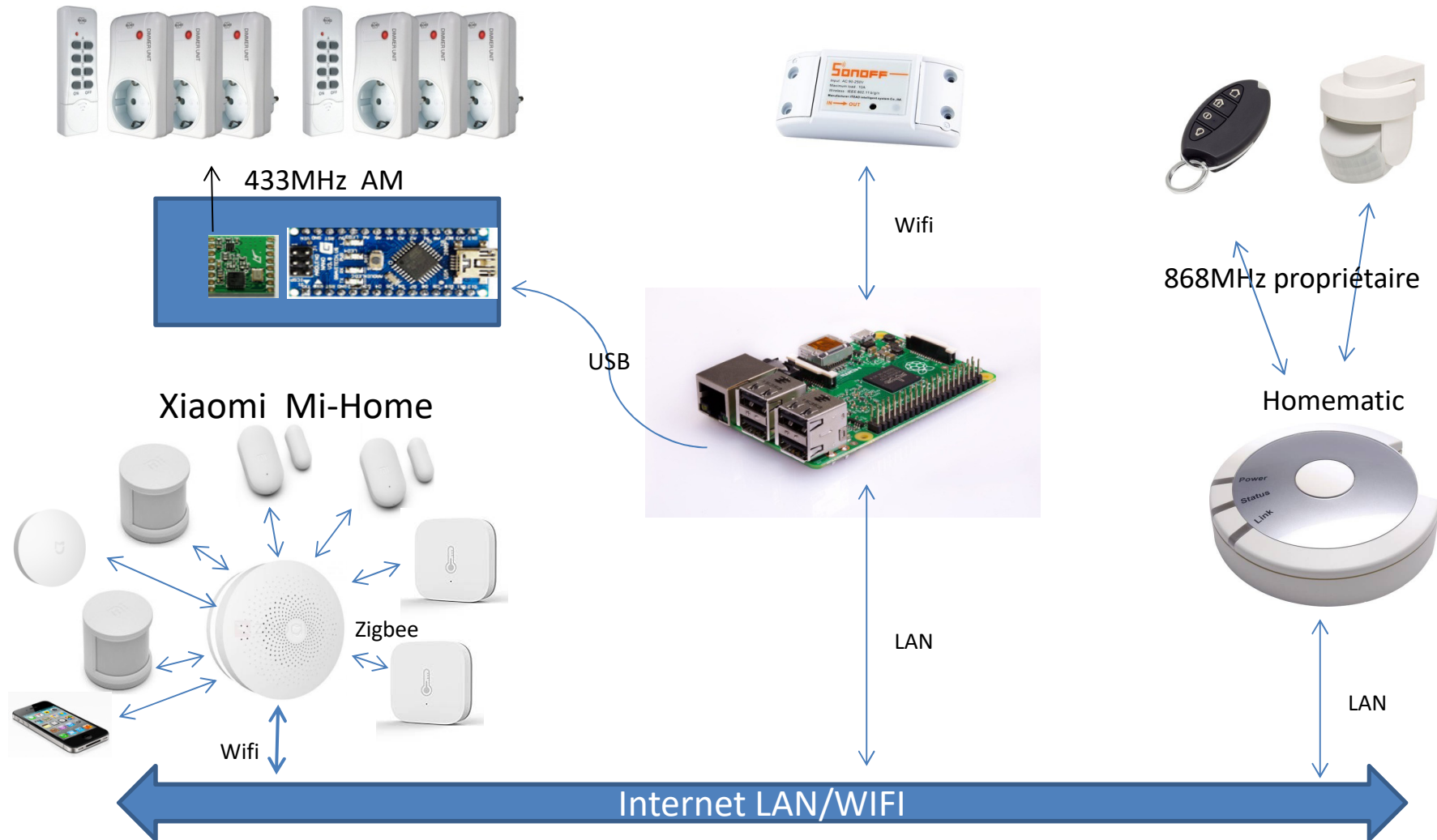
Configuration 2016



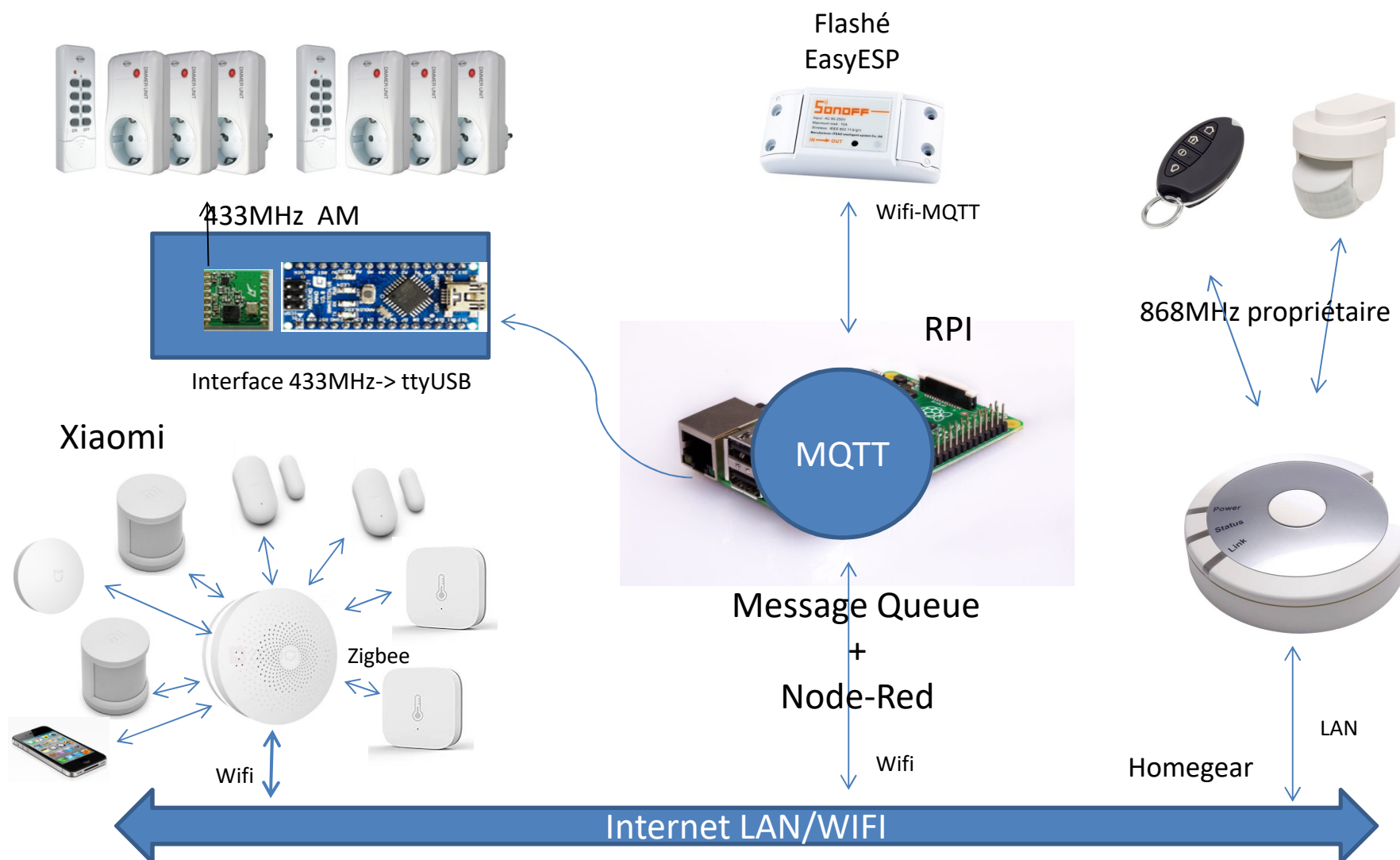
Elimination des composants propriétaires 2017



Installation actuelle



Installation 2018



Programme de la soirée

- Introduction
- **Xiaomi, capteurs domotique low-cost**
 - Installation de Xiaomi
 - Configuration de capteurs
 - Programmation de tâches
- Intégration de Xiaomi dans Node-Red
- A suivre:
 - Construction d'un client MQTT pour notre domotique

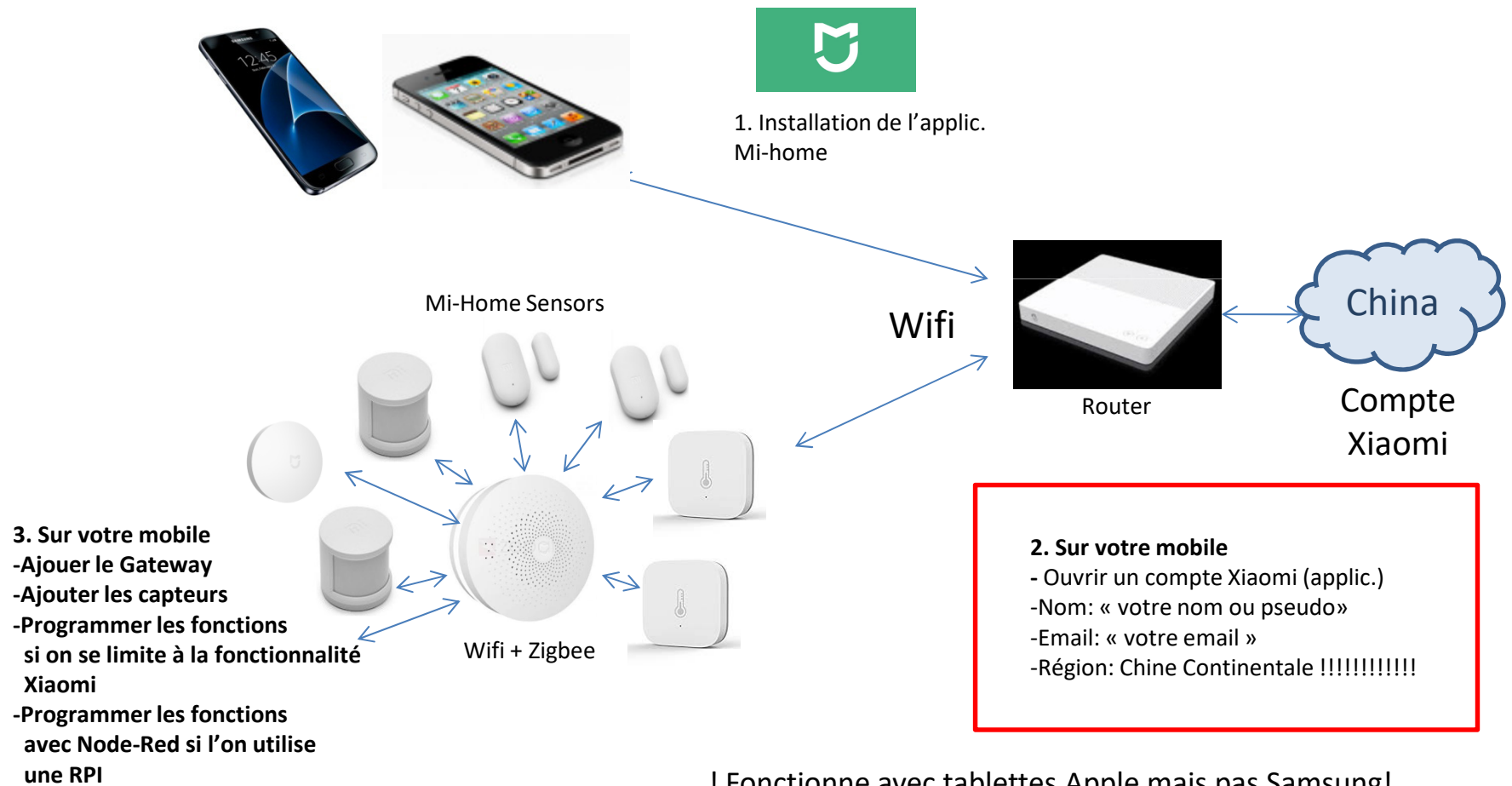
Xiaomi

Installation de base



Xiaomi

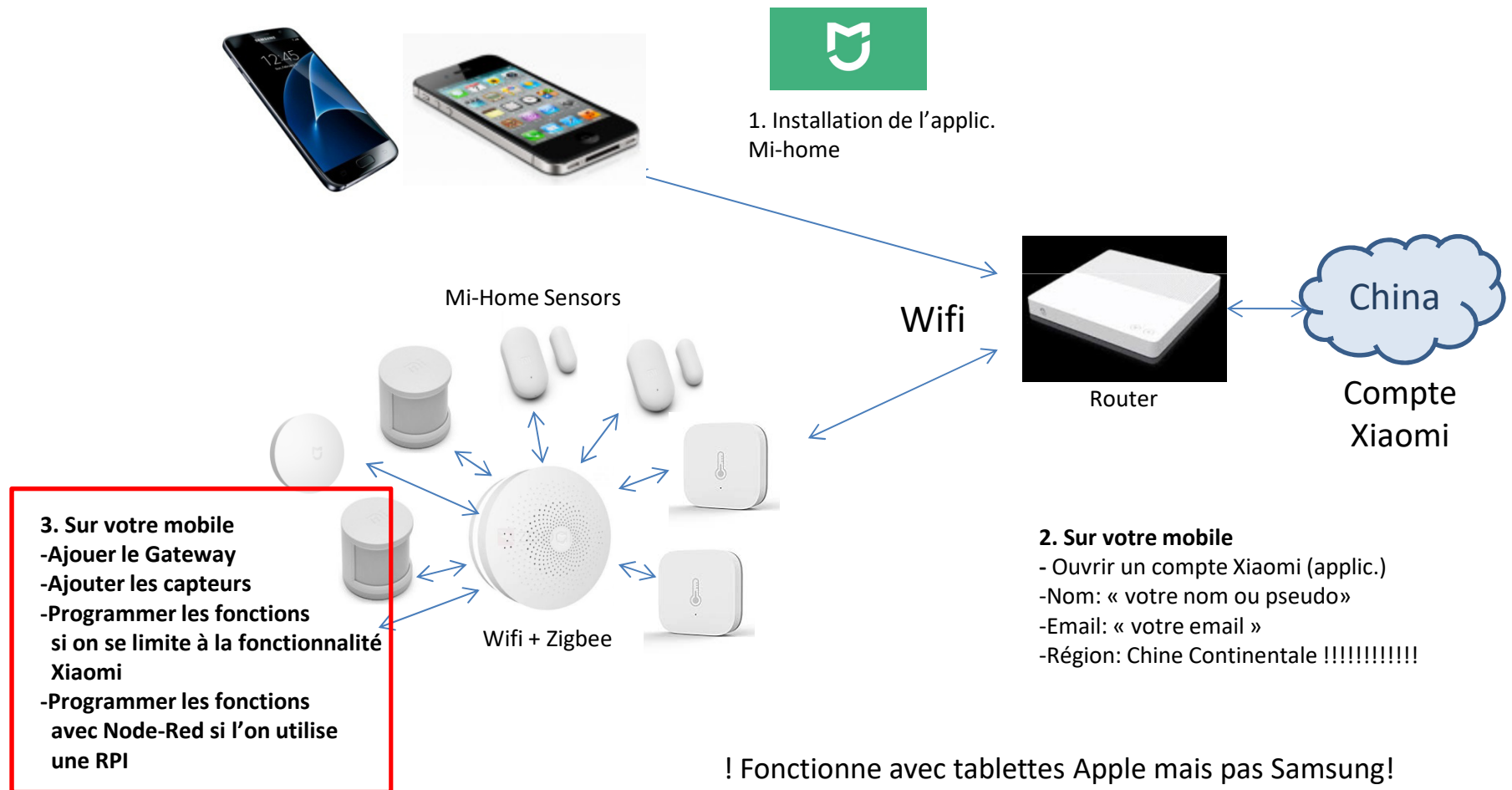
Installation de base



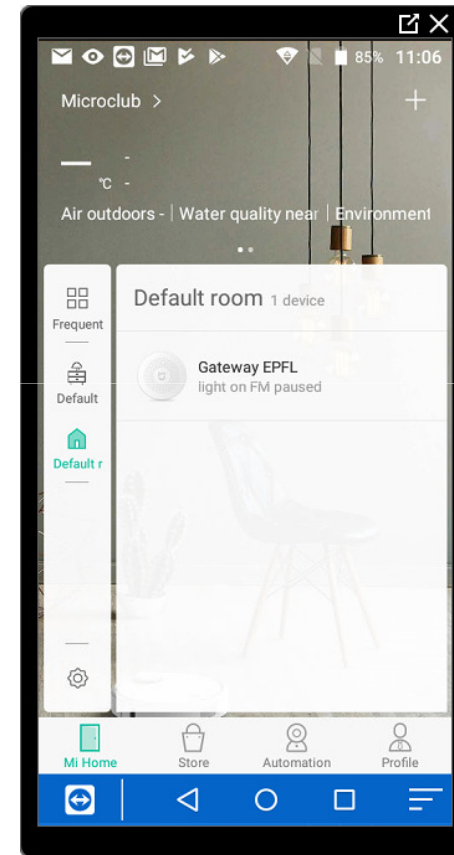
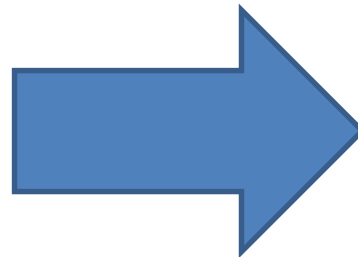
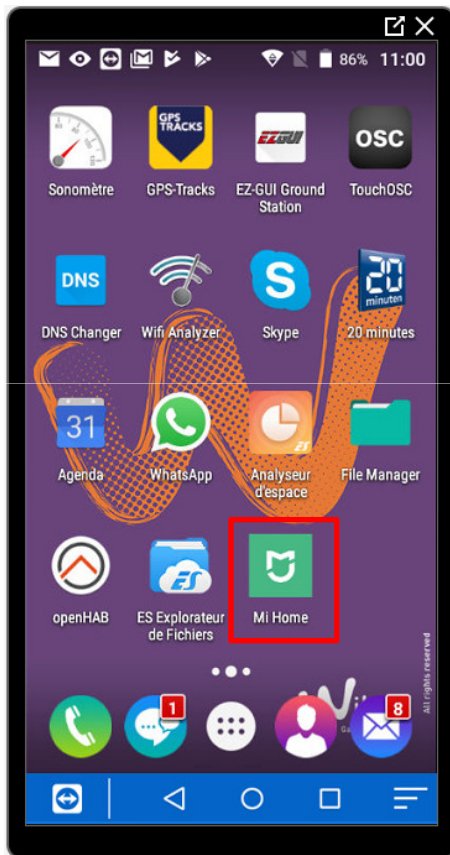
! Fonctionne avec tablettes Apple mais pas Samsung!

Xiaomi

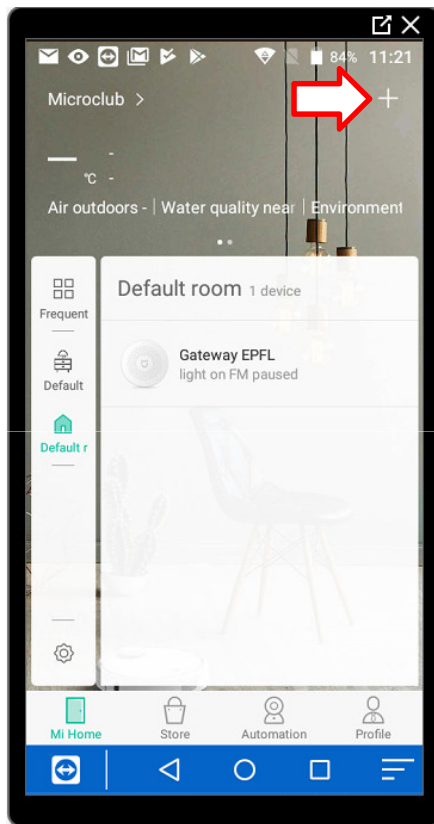
Installation de base



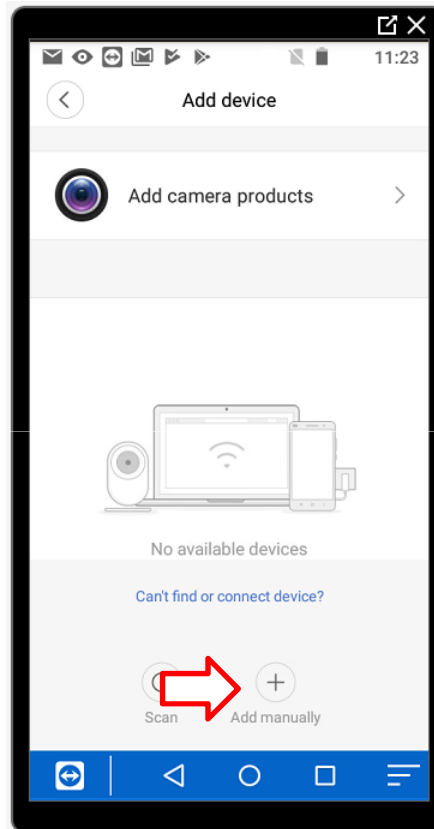
Détails capteurs (demo)



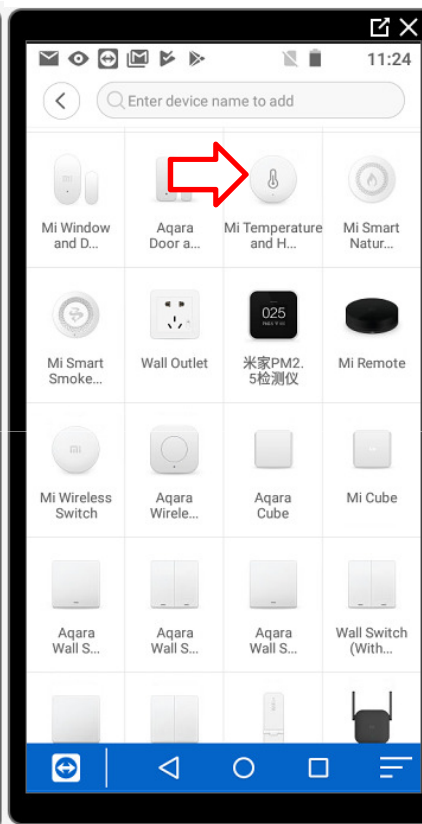
Ajouter les capteurs (1/2)



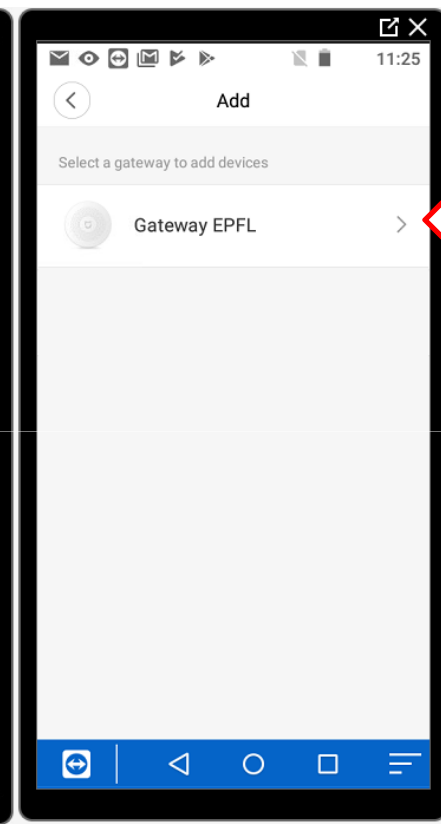
Cliquer sur « + »



Puis Add manually

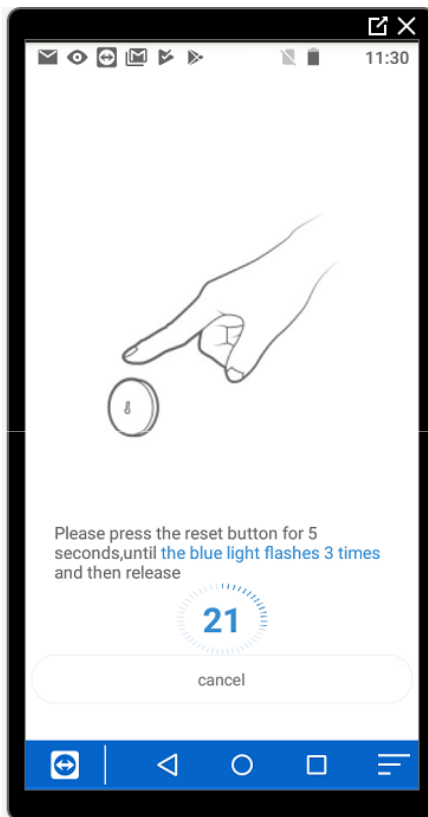


Chercher et sélectionner le capteur à ajouter

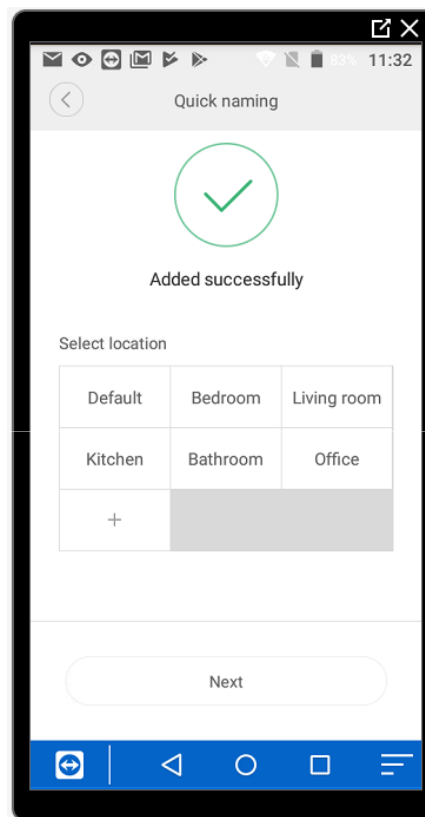


Choisir le GW
Préalablement
installé

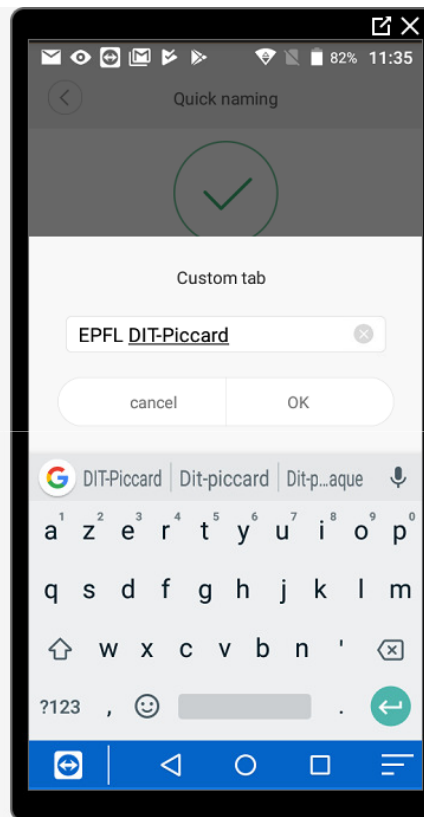
Ajouter les capteurs (2/2)



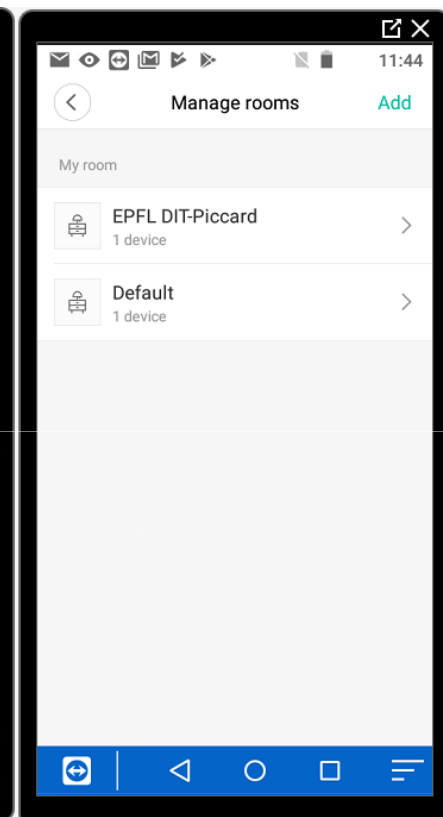
Identifier
le capteur



Appairage réussi
On choisit la pièce où
se trouve le capteur

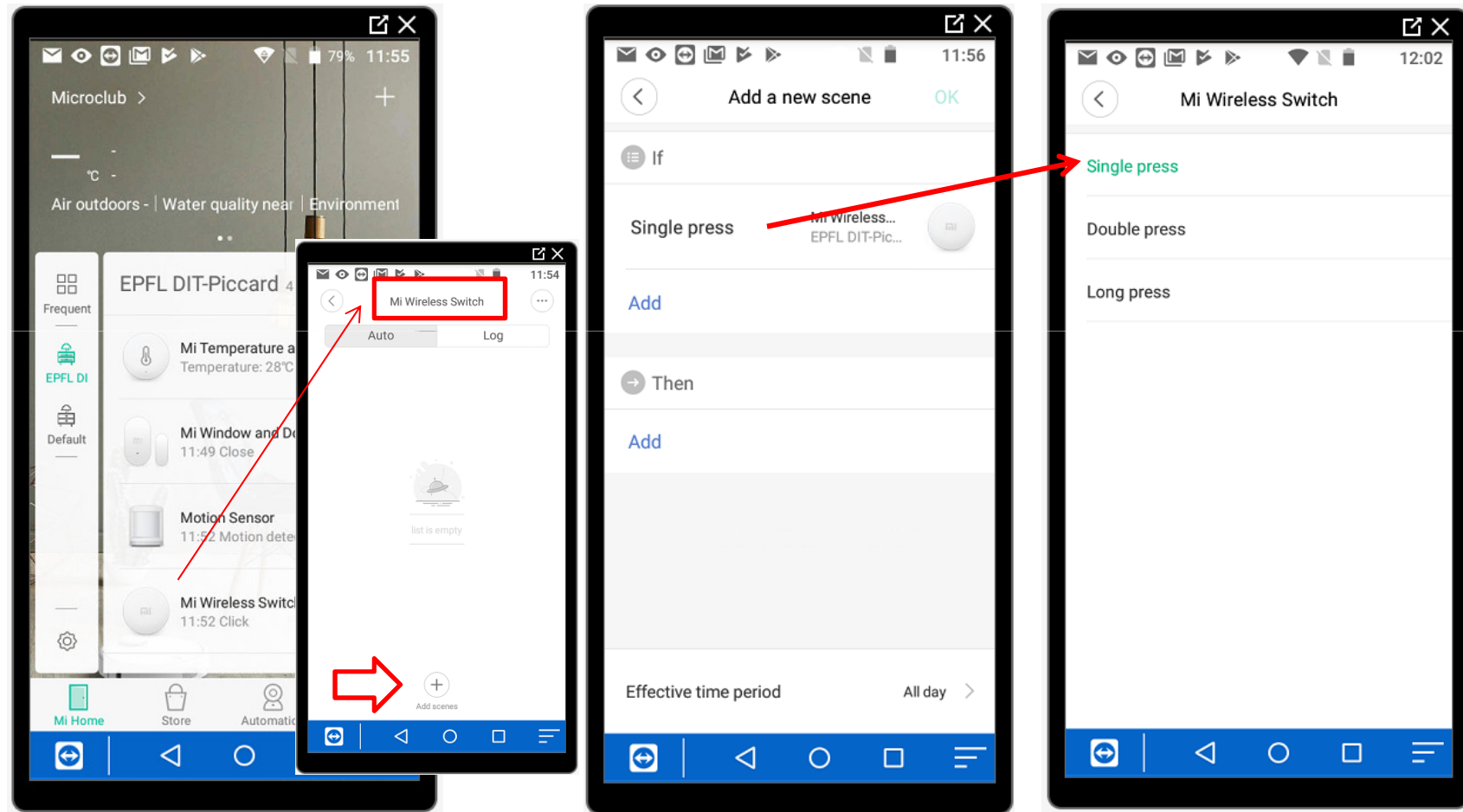


On peut ajouter
un nom de pièce



On ajoute les
autres capteurs
de la même
manière

Configuration de tâches (scenes) (sur Xiaomi)



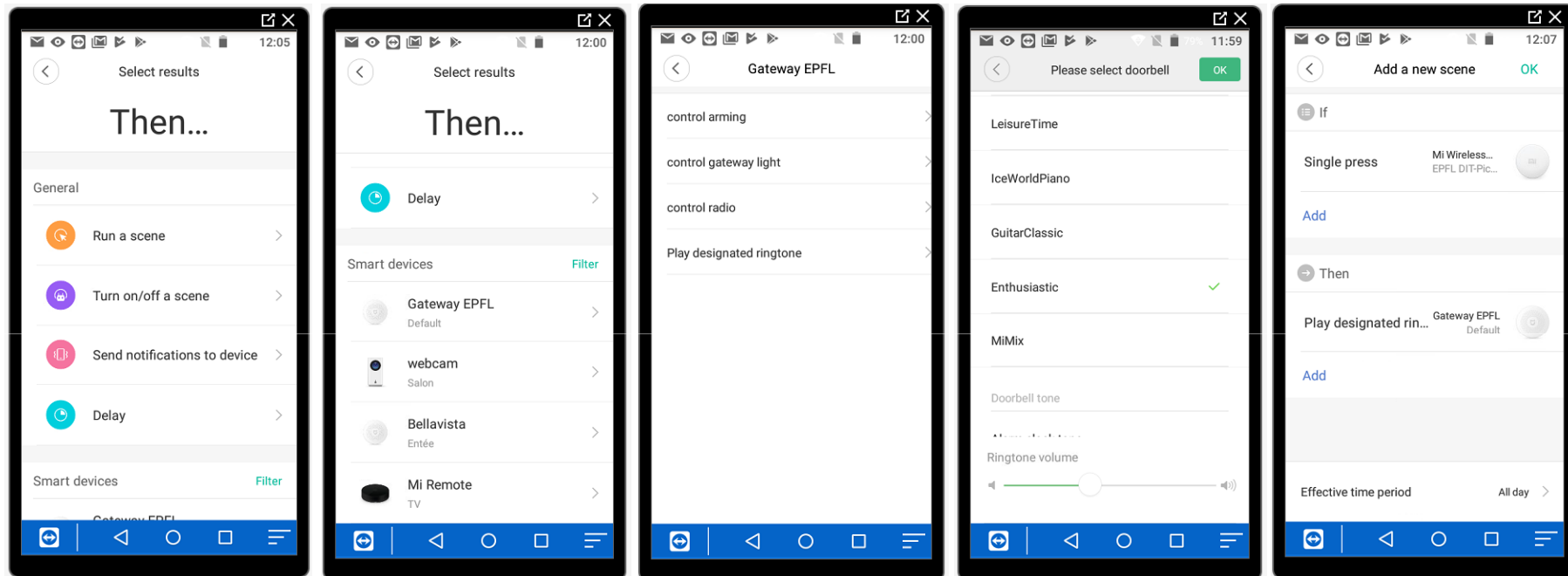
Sélectionner
un capteur

Ajouter une
Scène

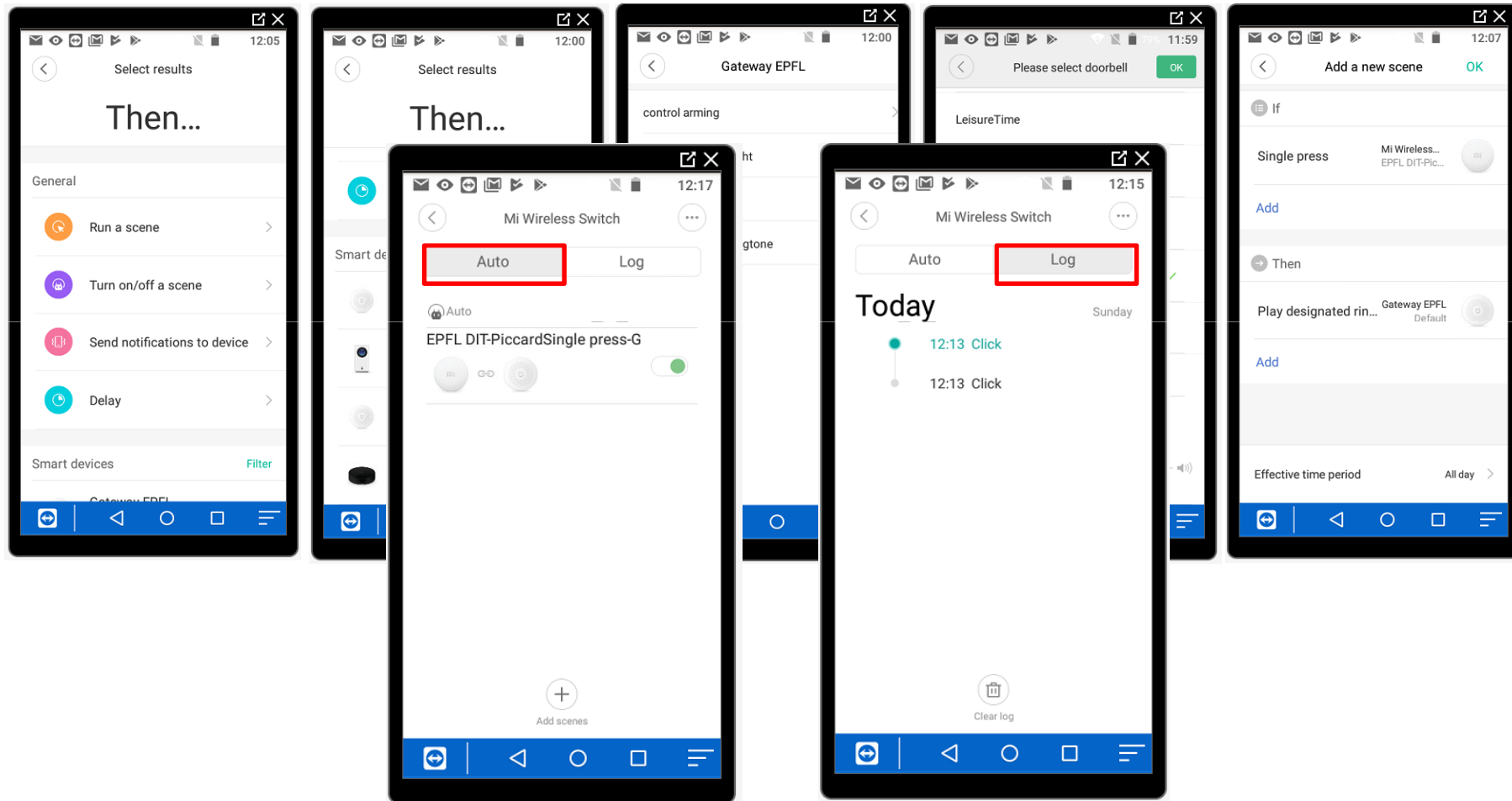
Rolf Ziegler 5/2018

Configurer une action
Type: if..... then.....

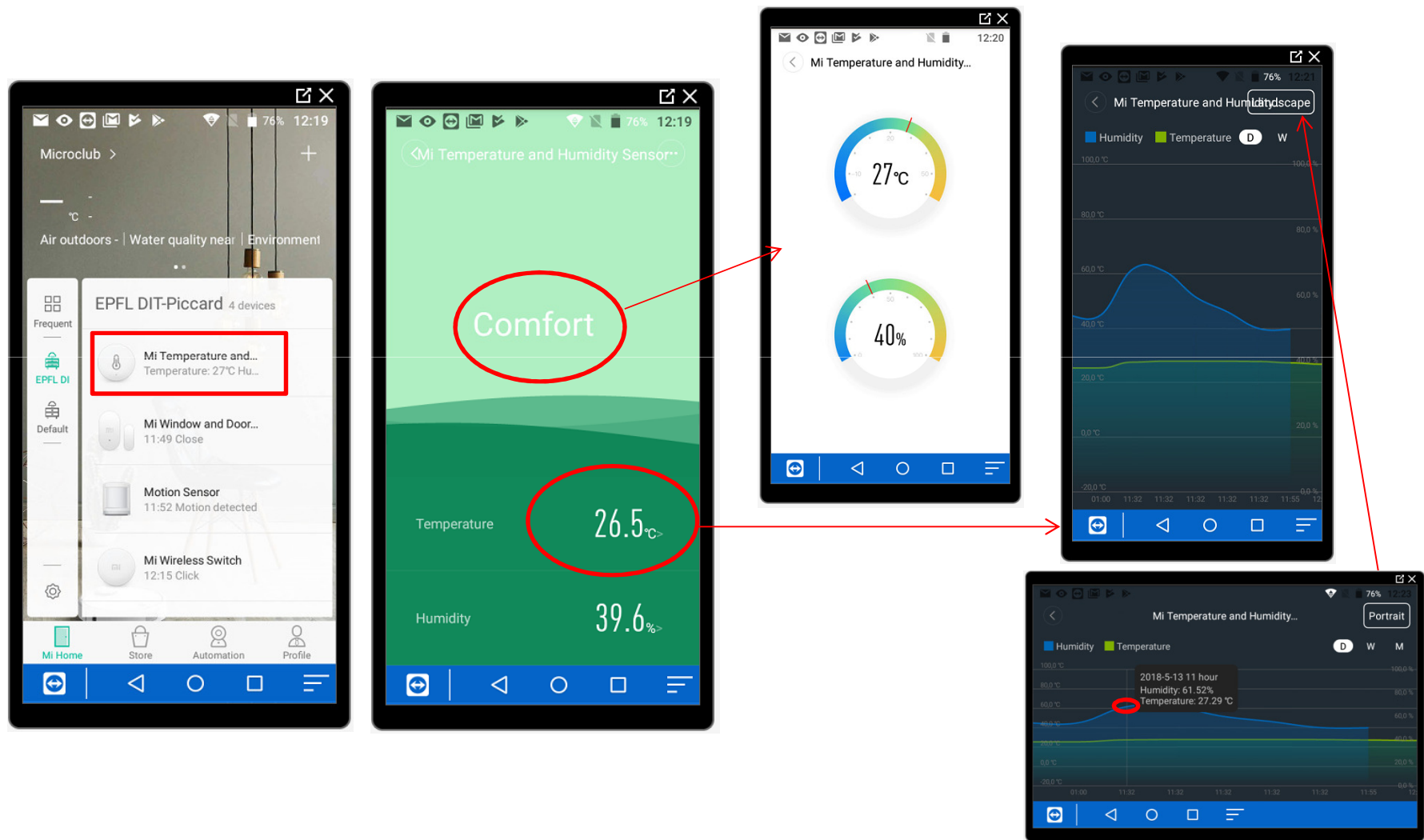
Configuration de tâche if---then



Configuration de tâche if---then



Exemple: Thermomètre



Résumé Xiaomi

- Capteurs bon marché
- Grand choix de capteurs, même camera/aspir.
- Système autonome (par iPhone/Smartphone)
- Bonne fonctionnalité
- Accessible par le Cloud (tel. Portable) en tout temps
- Envoi d'alarmes, notifications en fonction des tâches configurées
- Manque des interrupteurs EU/CH

Programme de la soirée (suite)

- Xiaomi, capteurs domotique low-cost
- **Intégration de Xiaomi dans Node-Red**
 - Installation de « Openhabian » (pack RPI)
 - Activation de MQTT, Homegear et Node-Red
 - Programmation sous Node-Red
 - Liaison avec Xiaomi
- A suivre:
 - Construction d'un client MQTT pour notre domotique

Débuts avec Openhab2.0

www.openhab.org

1^{ère} installation manuelle sous linux, tous les composants à installer séparément

The screenshot shows the OpenHAB website's 'Downloads' page. The navigation bar includes links for Home, Downloads (active), Documentation, Community, Github, and myopenHAB. The main heading is 'Downloads' with a breadcrumb 'HOME / DOWNLOADS'. Below this, the section 'openHAB Distribution 2.2' is displayed. A paragraph explains that these are packages for download and self-installation on any platform supporting Java (Linux, macOS, Windows). It also mentions other installation options and refers to the 'installation overview' in the documentation. Three download options are presented: 'Runtime Distribution' (a zipped version of openHAB runtime), 'Add-ons' (all openHAB add-ons for offline installation), and 'Legacy Add-ons' (1.x versions of add-ons). Each option has a 'Download' button. Below the download options is a section titled 'Featured Technologies' which displays four technology logos and descriptions: KNX (worldwide standard for home automation), EnOcean (energy harvesting wireless technology), INSTEON (powerline based networking technology), and HomeMatic (wireless home automation system by eQ-3 AG).

openHAB

Home Downloads Documentation Community Github myopenHAB

Downloads

HOME / DOWNLOADS

openHAB Distribution 2.2

These are package for download and self-installation on any platform that supports java, i.e. it can be used on Linux, macOS and Windows likewise. Besides this package, there are other installation options available, so please make sure to check our [installation overview](#) in the documentation first.

Runtime Distribution

This is a zipped version of openHAB runtime. It is configured to install any add-on from a remote repository.

[Download](#)

Add-ons

This package includes all openHAB add-ons for optional offline installation.

[Download](#)

Legacy Add-ons

This package includes 1.x versions of add-ons, for which there is already a 2.x alternative available

Featured Technologies

KNX

KNX is the worldwide STANDARD for all applications in home automation and building control

EnOcean

EnOcean is an energy harvesting wireless technology used primarily in building automation systems

Insteon

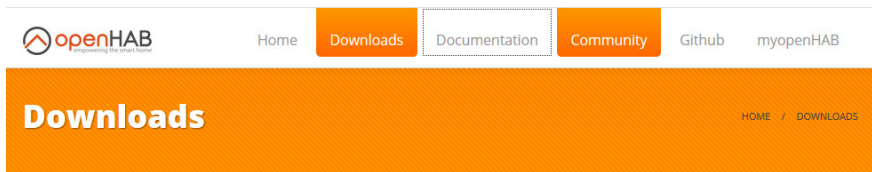
INSTEON is a powerline based networking technology for the connected home..

HomeMatic

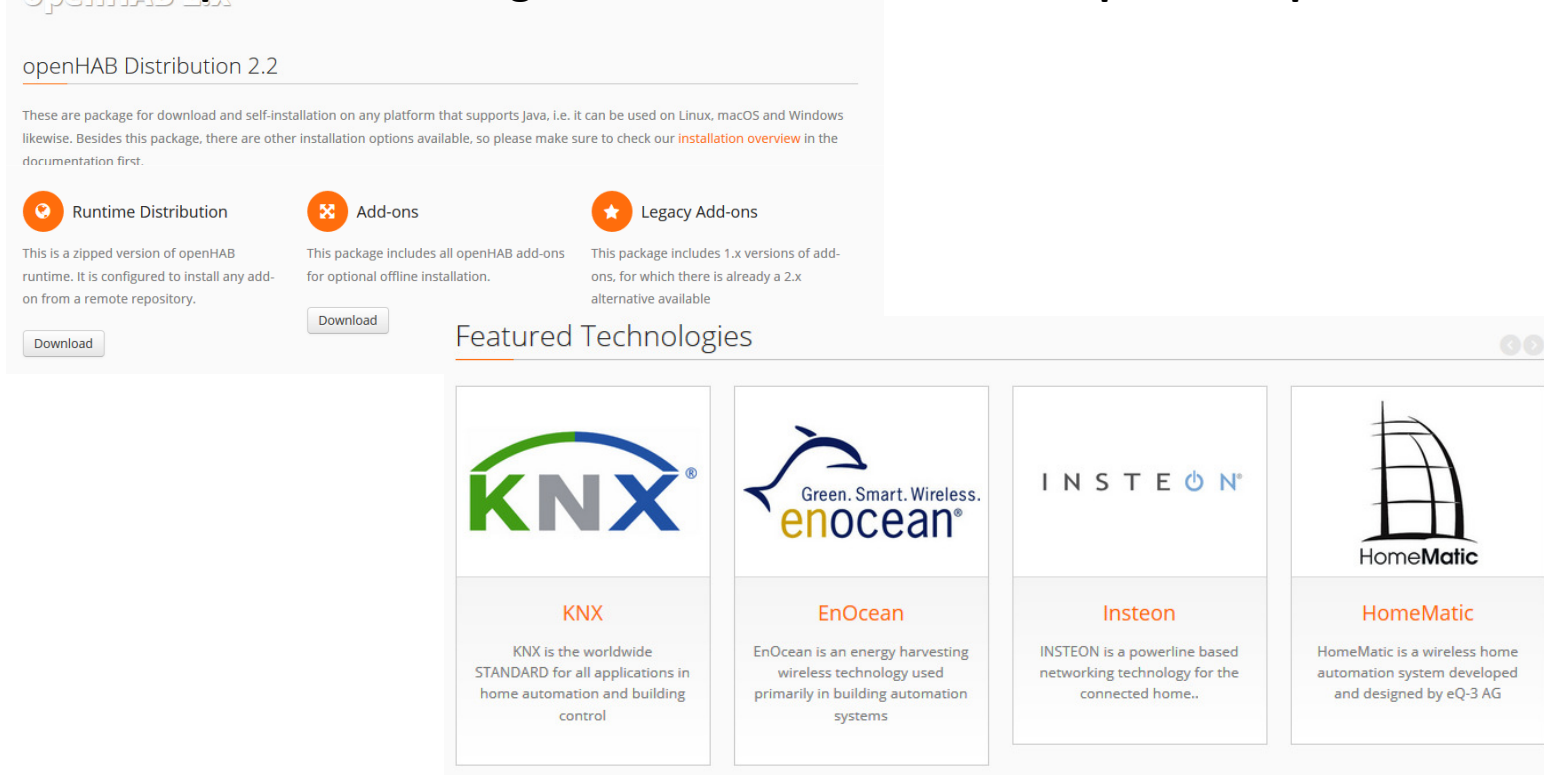
HomeMatic is a wireless home automation system developed and designed by eQ-3 AG

Débuts avec Openhab2.0

www.openhab.org



Nouveau: Openhabian, Image à flasher sur SD contenant plusieurs produits

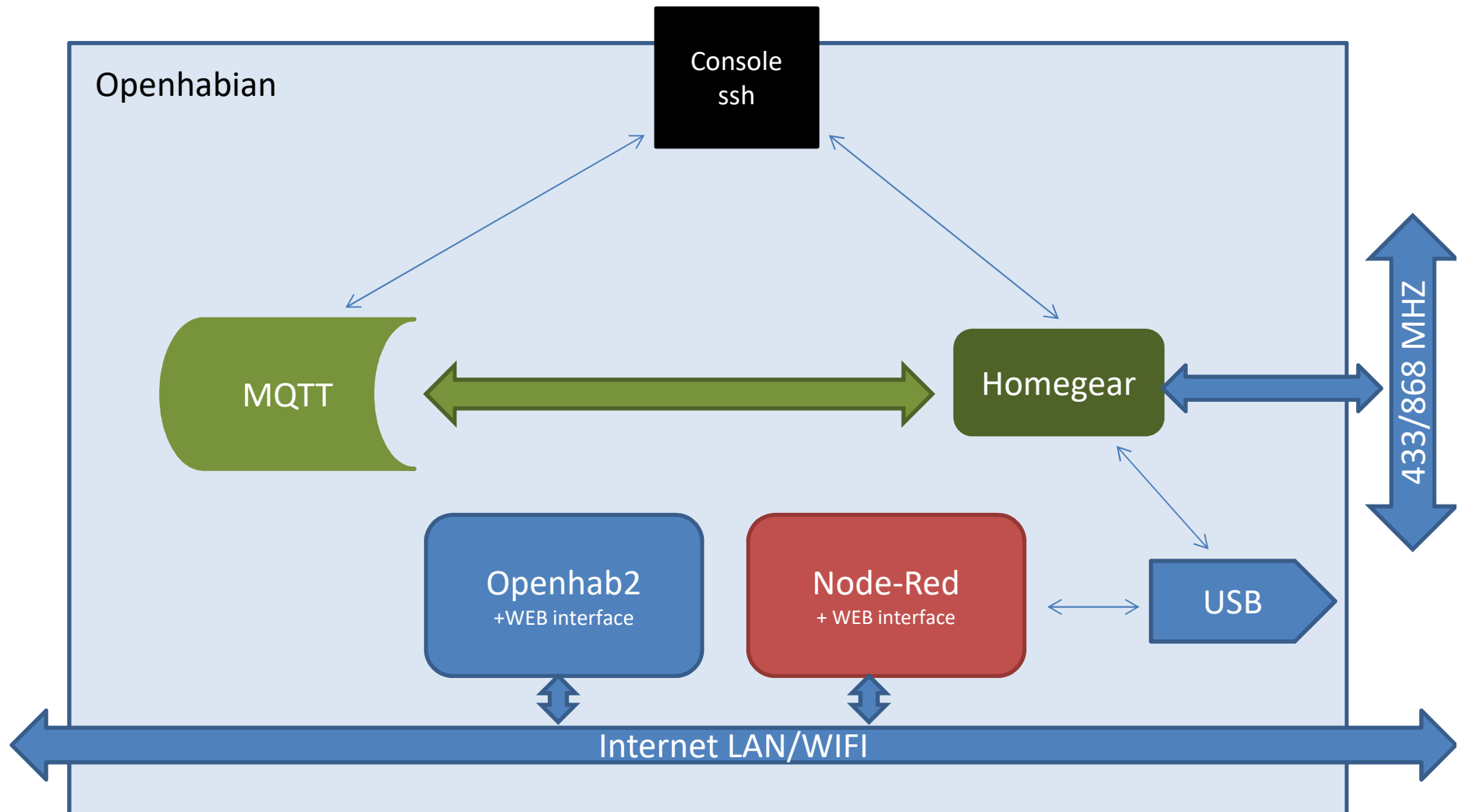


Openhabian, firmware RPI

Collection de logiciels préinstallés

- Container linux à flasher sur carte SD pour RPI
 - Contient tous les éléments pour notre domotique
 - **Openhab2** (module domotique Open-Source) Contient des dizaines d'interfaces de produits Domotiques /Capteurs etc.
 - **Mosquito, MQTT** (Messagerie FIFO, centre de communication)
 - **Node-Red** (module de programmation graphique avec UI)
 - **Homegear** (Interface EnOcean +Homematic + Intertechno +KNX + Nanoleaf + Sonos)
 - Tous les produits peuvent communiquer par MQTT avec Openhab2 et/ou Node-Red

Communication



Installation sur carte SD

- Télécharger le container de GitHub [openhab/openhabian](https://github.com/openhab/openhabian)
- Flash la carte SD (16GB) avec « ETCHER » ou similaire
- Connection SSH avec PUTTY
login: openhabian Password: openhabian
- Démarrer le configurateur avec: « sudo openhabian-config »
- Installation des logiciels

Openhabian-config

- Permet d'installer tous les logiciels de la présentation à l'exception de XIAOMI

```
Welcome to the openHABian Configuration Tool [master]v1.4.1-393(384d552)

00 | About openHABian      Information about the openHABian project and this tool

01 | Update                Pull the latest revision of the openHABian Configuration Tool
02 | Upgrade System        Upgrade all installed software packages to their newest version
03 | openHAB 2.2.0 stable  Switch from openHAB 2.1 or 2.2-snapshot to the 2.2 stable release

10 | Apply Improvements    Apply the latest improvements to the basic openHABian setup ►
20 | Optional Components   Choose from a set of optional software components ►
30 | System Settings       A range of system and hardware related configuration steps ►
40 | openHAB related        Switch the installed openHAB version or apply tweaks ►
50 | Backup/Restore        Manage backups and restore your system ►
60 | Manual/Fresh Setup    Go through all openHABian setup steps manually ►

99 | Help                  Further options and guidance with Linux and openHAB

<Execute>                  <Exit>
```

Installation des logiciels par interface SSH

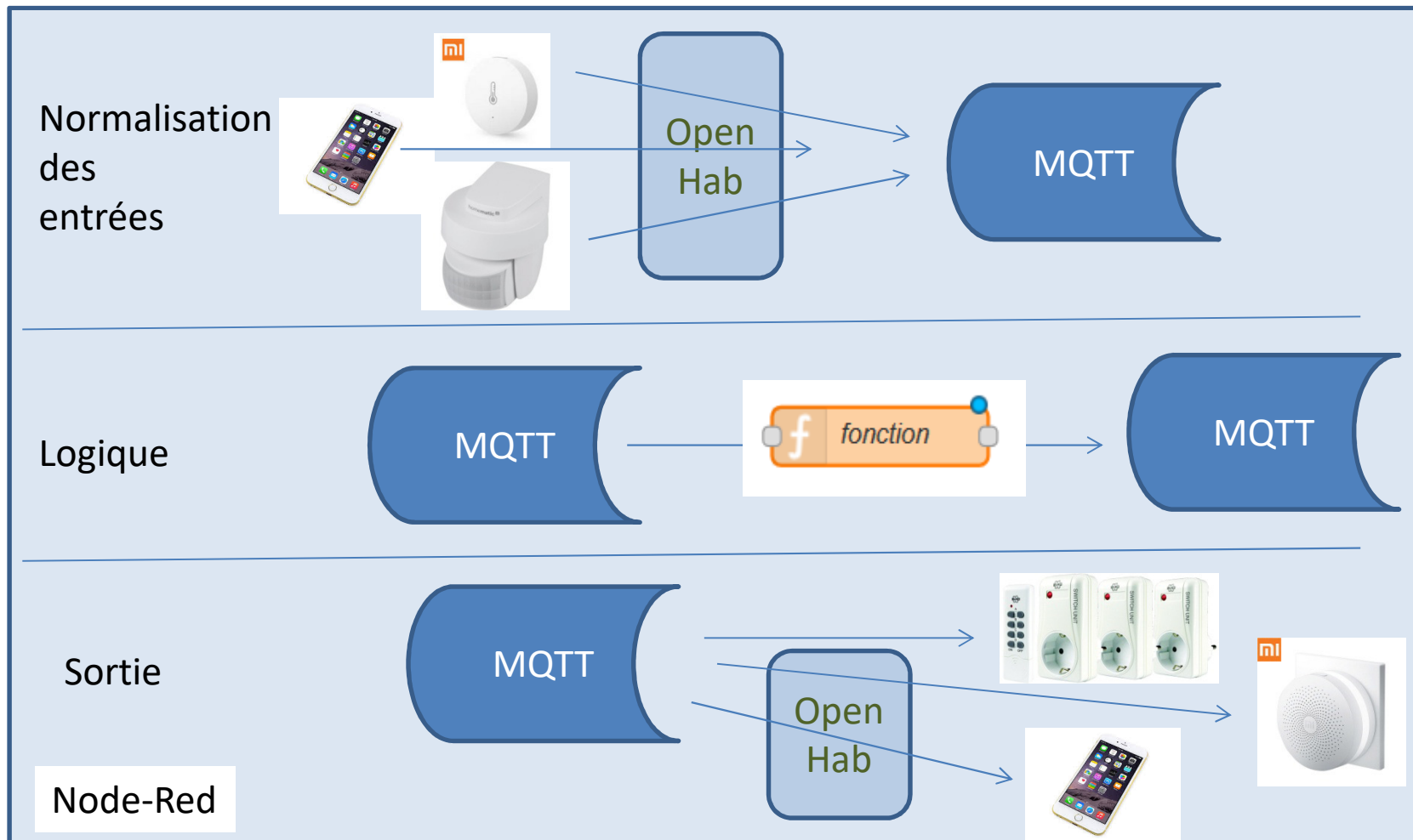
- Openhab2.0 n'est pas obligatoirement nécessaire si l'on suit les modules de ma démo
- 01 update
- 02 optional components
 - 22 Mosquitto (MQTT)
 - 25 Node-Red (programmation de la logique)
 - 26 Home-Gear (Interface Homematic etc)
- 03 system settings
 - Langue, time-zone
 - Changer le mot de passe « root »
 - Connecter la RPI sur votre WIFI local

Configuration des modules

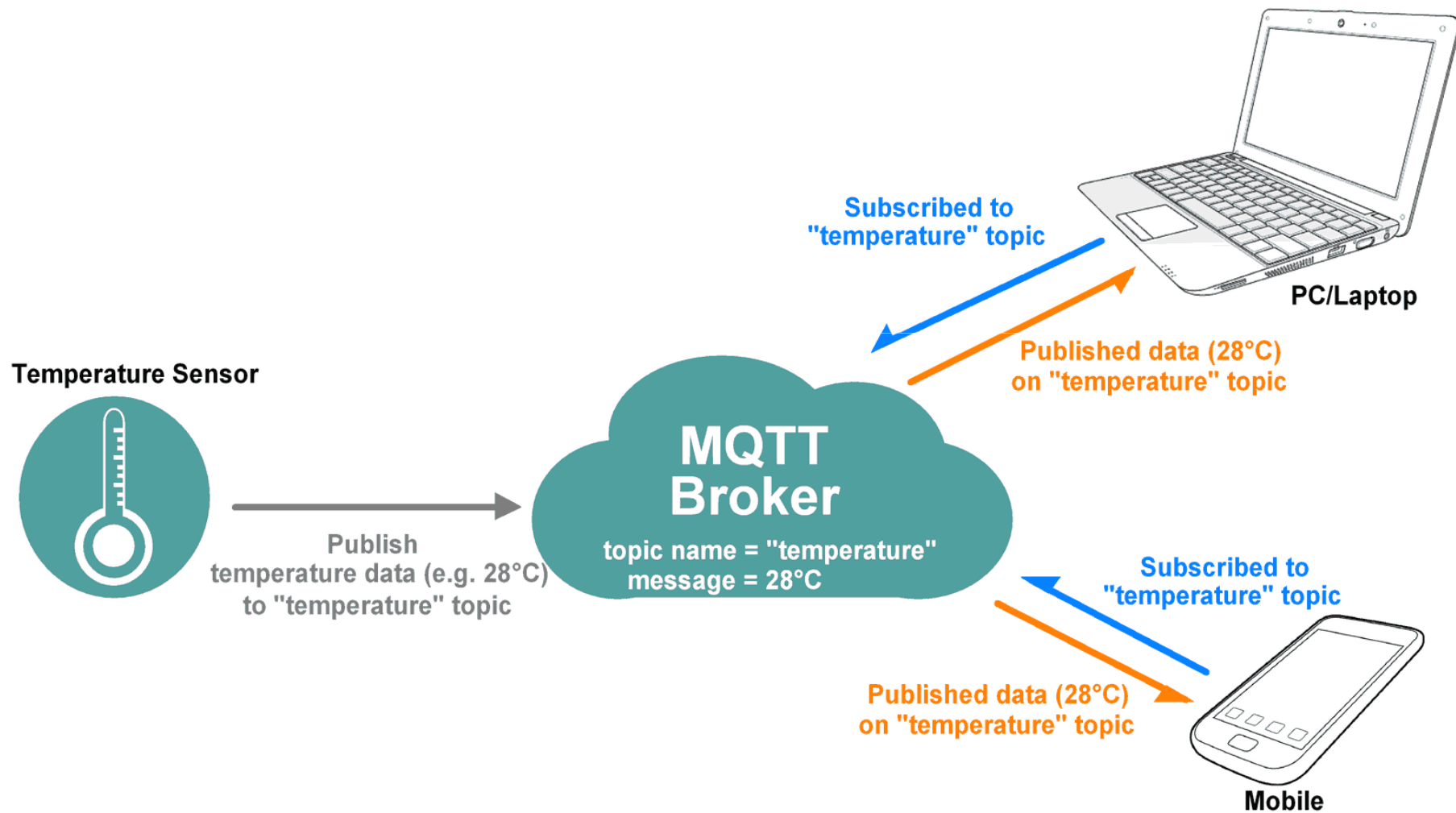
- La configuration du serveur MQTT se fait en éditant le fichier `/etc/mosquitto/mqtt.conf`
- Node-Red est atteignable sous `<RPI-IP:1880>`
 - La configuration et l'installation de modules se fait dans node-red même (démarrer l'interface)
- L'interface utilisateur N-R sous `<RPI-PI:1880/ui>`

Node-Red + MQTT

Mon principe de communication



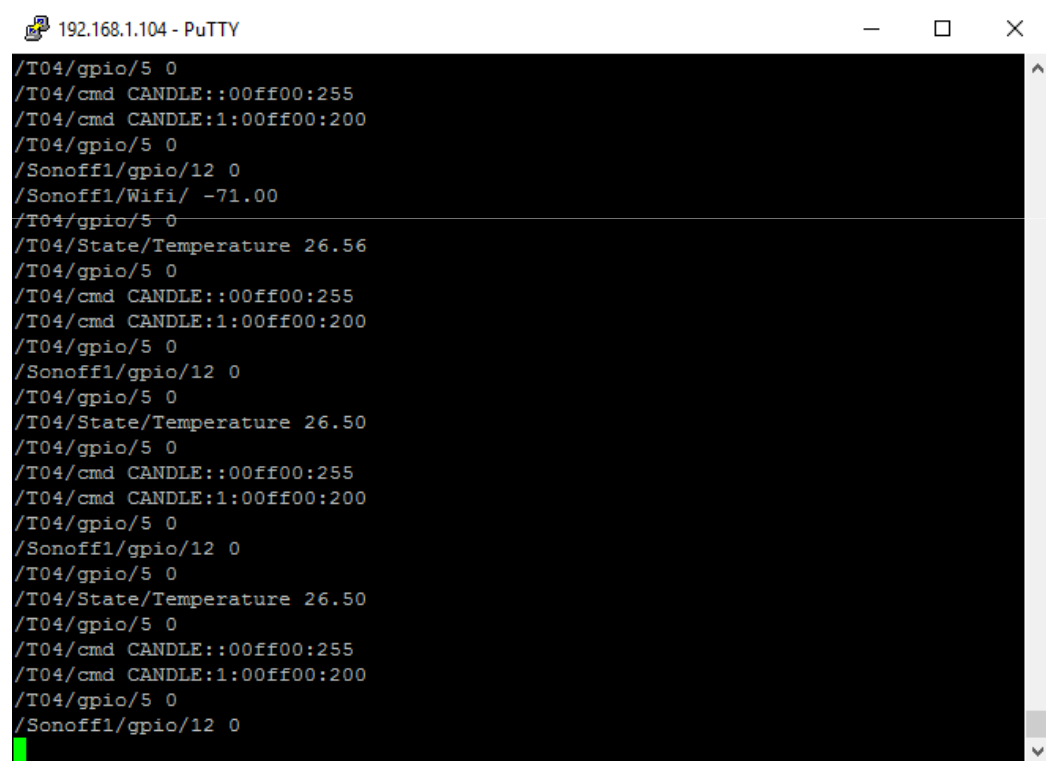
Fonctions MQTT rappel



MQTT Server log

- Utilisation du script mqtt.sh avec «./mqtt.sh »
contient: mosquitto_sub -v -t /#

MQTT n'a pas
d'interface
HTML !



A screenshot of a PuTTY terminal window titled '192.168.1.104 - PuTTY'. The terminal displays a series of MQTT messages received by a subscriber. The messages are formatted as topic/path payload, where the payload is a JSON object. The topics include temperature readings, candle commands, and GPIO pin states. The messages are as follows:

```
/T04/gpio/5 0
/T04/cmd CANDLE::00ff00:255
/T04/cmd CANDLE:1:00ff00:200
/T04/gpio/5 0
/Sonoff1/gpio/12 0
/Sonoff1/Wifi/ -71.00
/T04/gpio/5 0
/T04/State/Temperature 26.56
/T04/gpio/5 0
/T04/cmd CANDLE::00ff00:255
/T04/cmd CANDLE:1:00ff00:200
/T04/gpio/5 0
/Sonoff1/gpio/12 0
/T04/gpio/5 0
/T04/State/Temperature 26.50
/T04/gpio/5 0
/T04/cmd CANDLE::00ff00:255
/T04/cmd CANDLE:1:00ff00:200
/T04/gpio/5 0
/Sonoff1/gpio/12 0
/T04/gpio/5 0
/T04/State/Temperature 26.50
/T04/gpio/5 0
/T04/cmd CANDLE::00ff00:255
/T04/cmd CANDLE:1:00ff00:200
/T04/gpio/5 0
/Sonoff1/gpio/12 0
```

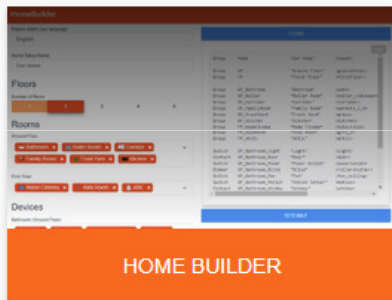
*ou sur windows avec MQTTfx

Openhab

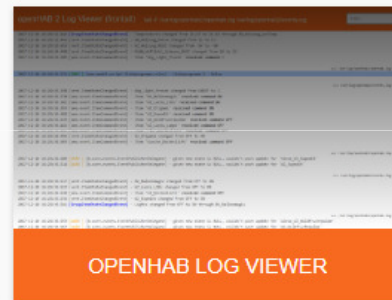
Interface matériel



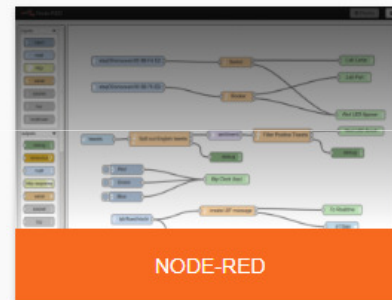
Welcome to openHAB 2



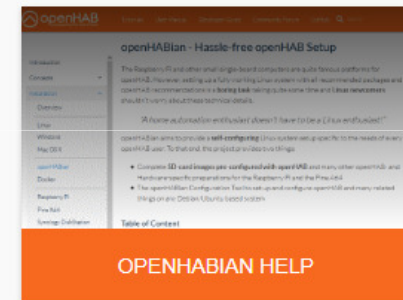
HOME BUILDER



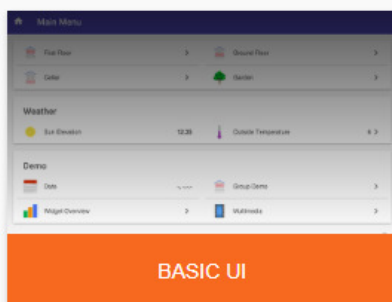
OPENHAB LOG VIEWER



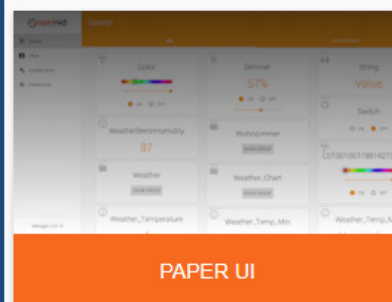
NODE-RED



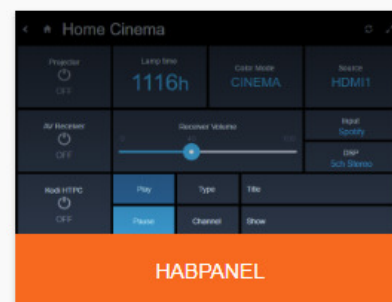
OPENHABIAN HELP



BASIC UI



PAPER UI



HABPANEL

Interfaces principaux

Getting started? Please refer to the [online documentation](#).

Interfaçage

Openhab ou directement Node-Red

The screenshot displays the openHAB 2 web interface. At the top, the openHAB logo and tagline "empowering the smart home" are visible. Below the header, a "Welcome to openHAB 2" message is shown. The main content area is divided into several sections:

- HOME BUILDER**: A section for configuring the home environment, showing a list of rooms and devices.
- BASIC UI**: A section for the basic user interface, showing a weather widget and a list of devices.
- PAPER UI**: A section for the Paper UI, showing a weather widget and a list of devices.
- OPENHAB LOG VIEWER**: A section for viewing the system logs.
- Navigation Menu**: A central menu with options: Control, Inbox, Configuration, Add-ons, and Preferences.
- Inbox**: A section displaying a list of discovered devices, including:
 - Alarm Speaker**: Wireless Indoor Siren, homematic:HG-HM-Sec-Sir-WM:D9B62456-3075-E8B5-7941-D8CF4DF5D808:OEQ0094793
 - Fumée Cuisine**: Wireless Smoke Detector, homematic:HG-HM-Sec-SD-2:ae77b013:NEQ0444431
 - Remote Rolf**: No Description, homematic:HG-HM-RC-Sec4-2:D9B62456-3075-E8B5-7941-D8CF4DF5D808:LEQ1313270
 - T-NEQ0444431**: Wireless Smoke Detector Team, homematic:HG-HM-Sec-SD-2-Team:D9B62456-3075-E8B5-7941-D8CF4DF5D808:T-NEQ0444431

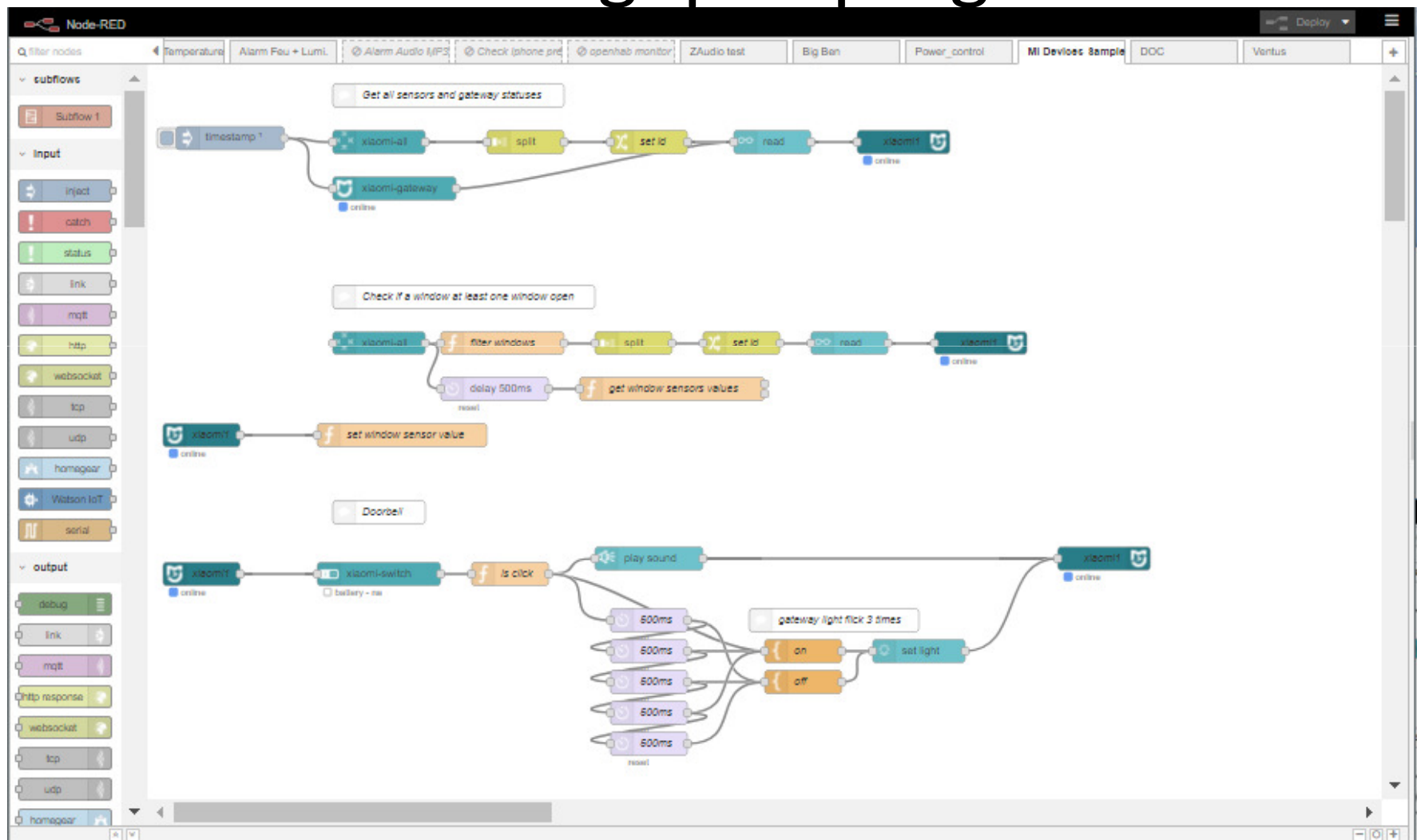
At the bottom of the inbox, there is a "SHOW IGNORED" button and a "Thing not listed? SEARCH FOR THINGS" link.

Interfaces matériel multiples tels que TV, Radio, etc (100 en tout)

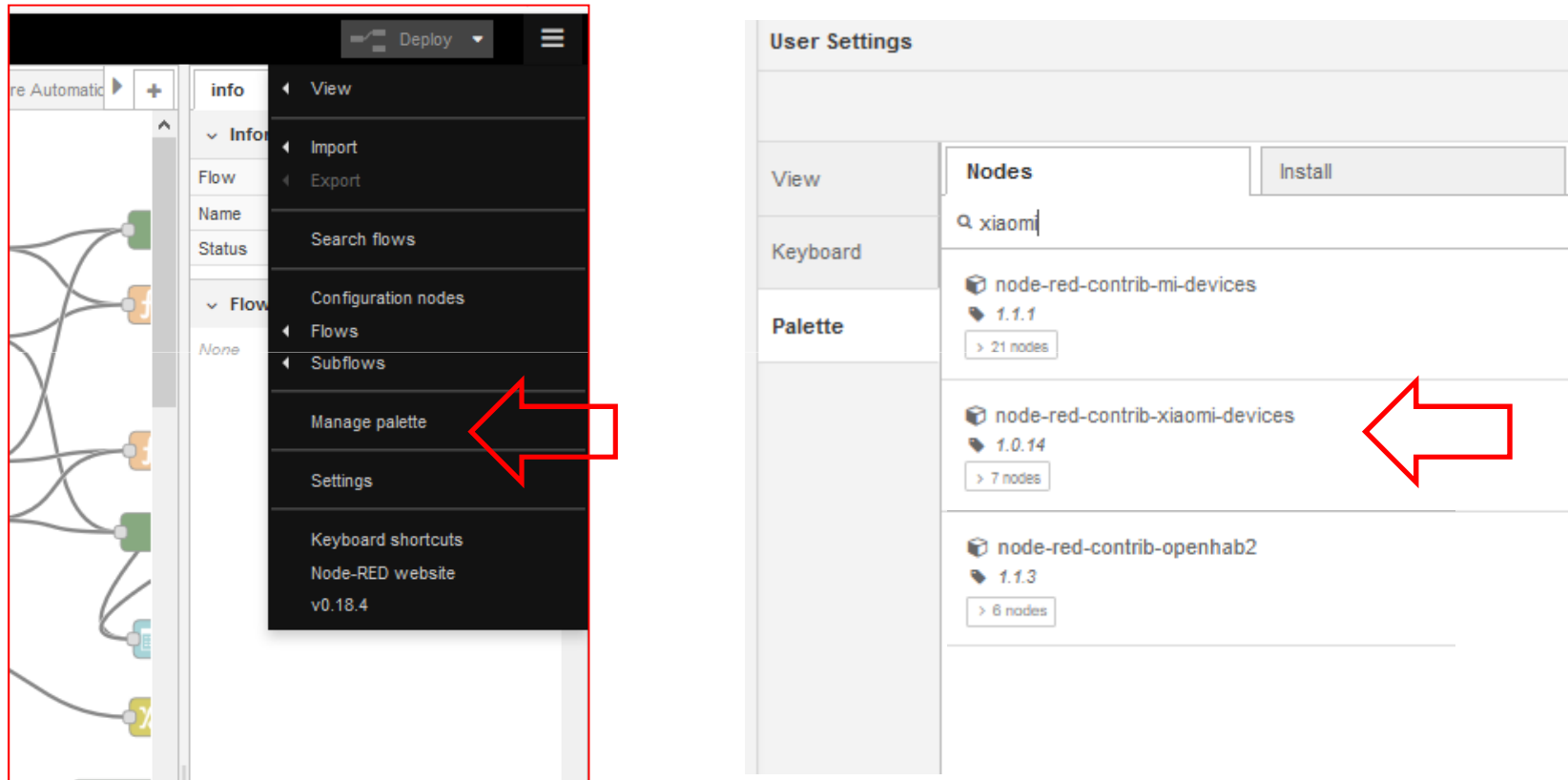
Rolf Ziegler 5/2018

Node-Red

Définition de logique+programmation

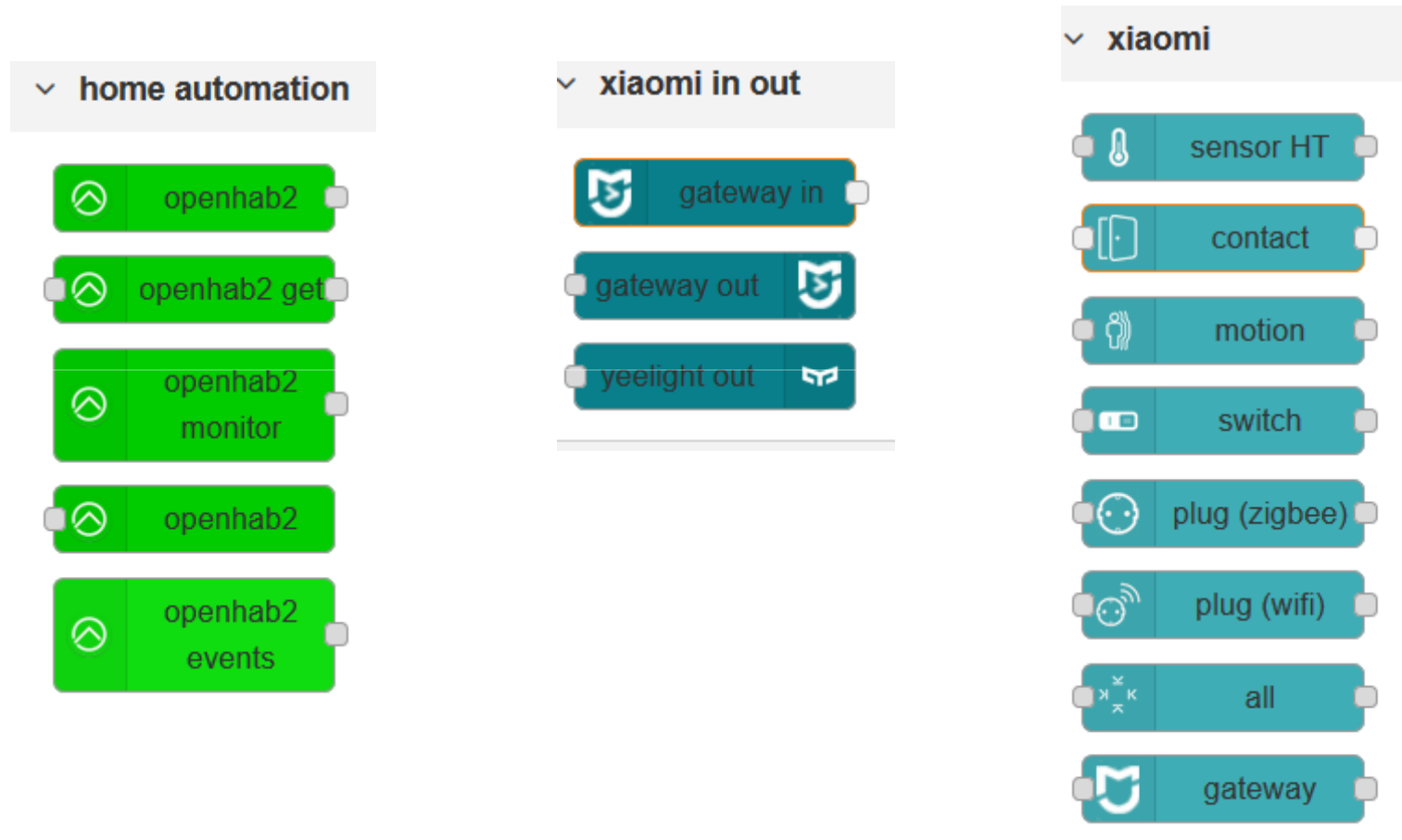


Interfaçage Xiaomi+Openhab avec Node-Red



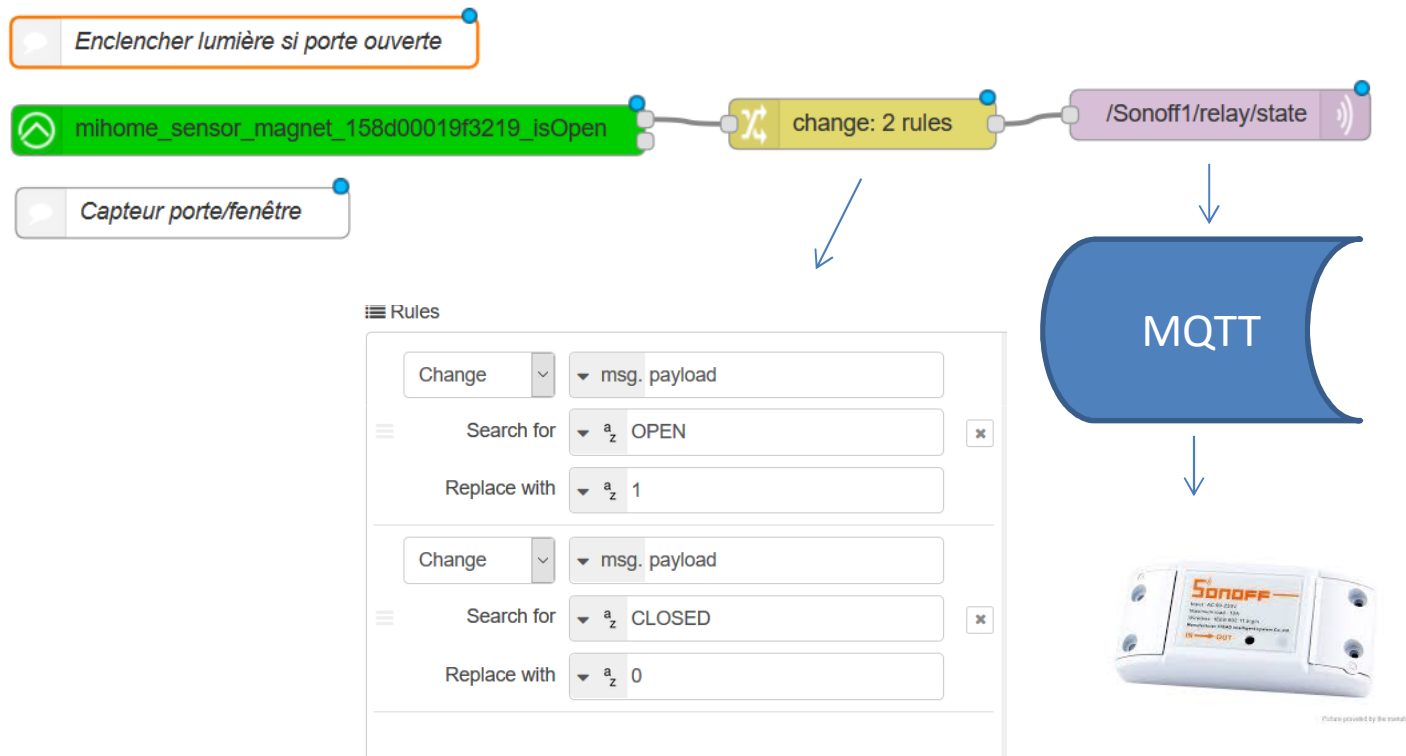
<https://github.com/pierrecle/node-red-contrib-mi-devices>

Exemple de programmation Xiaomi-Node-Red-MQTT-Sonoff

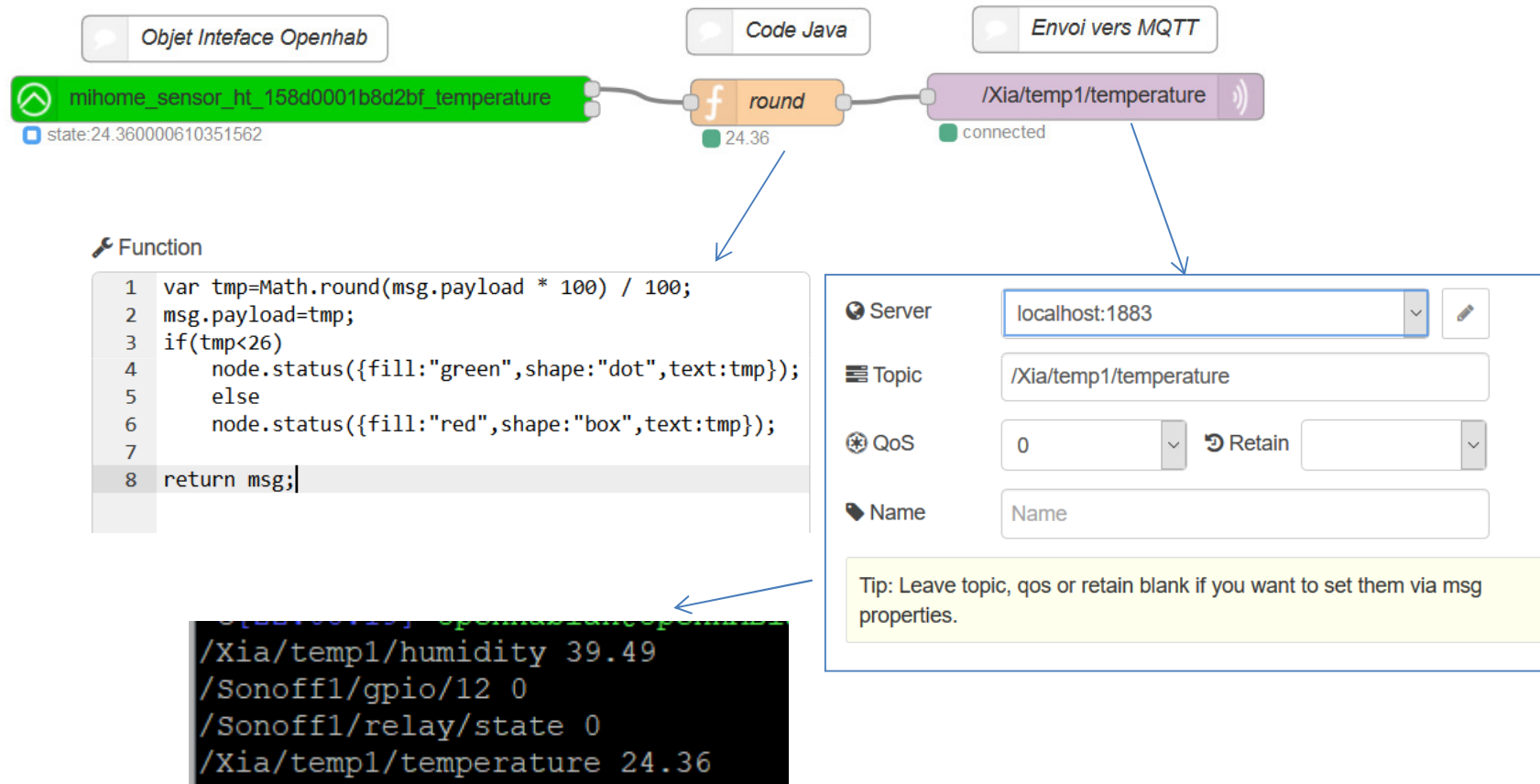


Exemple

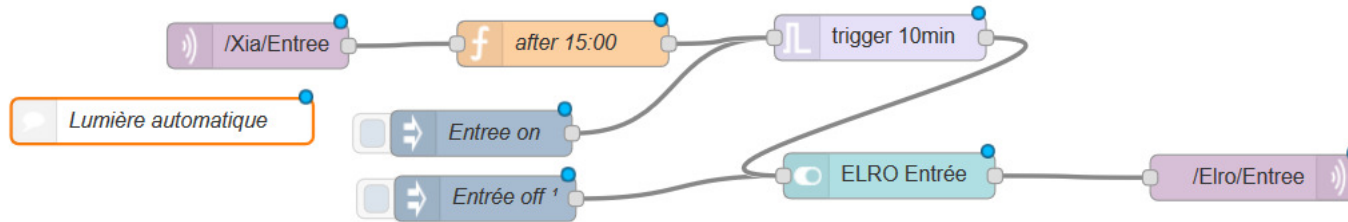
Allumer lumière avec Switch porte



Interface Openhabian Node-Red



Xiaomi mouvement allume la lumière pendant la nuit

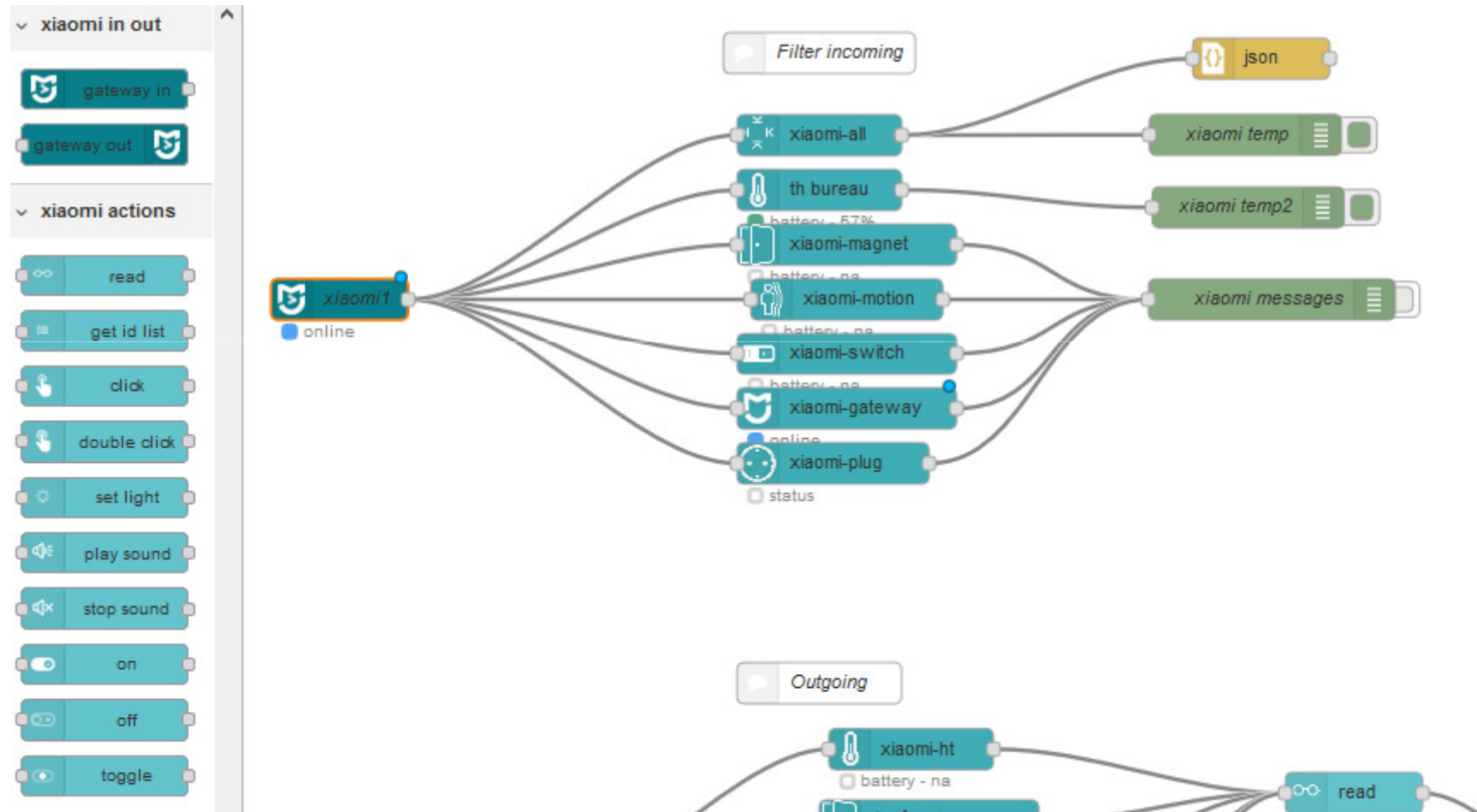


Function

```
1  var now = new Date();
2  var hour = now.getHours();
3  var lumiere = global.get("Lumiere_Jardin");
4  if((hour>15) || (hour <= 8) && (lumiere=="On" )){
5      if(msg.payload=="On")
6      {
7          node.status({fill:"green",shape:"dot",text:msg.payload});
8          msg.payload="1";
9      }
10     else
11     {
12         node.status({fill:"red",shape:"ring",text:msg.payload});
13         msg.payload="Off";
14     }
15     else payload="Off";
16     return msg;
17 }
```

Programmation en Java, ici par exemple enclenchement de lumière
Sur détection à l'entrée de l'appartement + que de 15h à 8h du matin

Xiaomi dans Node-Red



Configuration des capteurs sous Node-Red

Filter incoming

Delete Cancel Update

Name xiaomi1

IP address (v4 or v6) 192.168.1.123

SID (optional) sid

Key/Password 1612DD0D8EFA4AB0

158d0001a36565
th bureau

158d0001b7b7a2
fenêtre bureau

158d00015b1b4d
Motion Bureau

Outgoing

Clef d'identification du GW

Voir procédure dans la suit

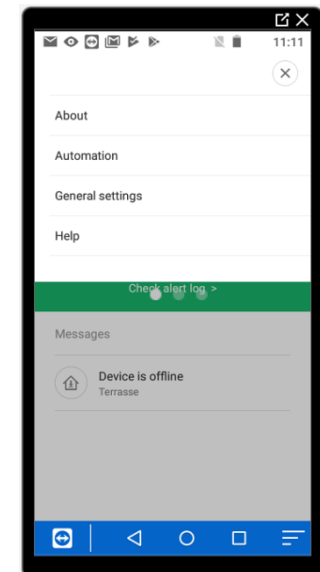
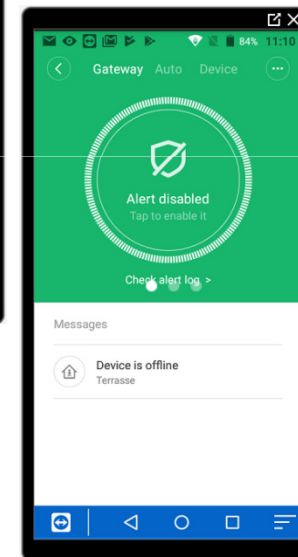
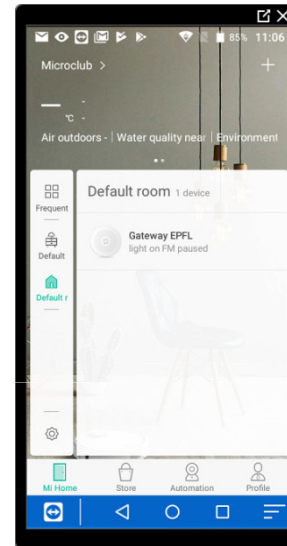
Identifiant des capteurs

Interface Xiaomi direct

- Eviterait d'utiliser Openhab2
- Théoriquement plus simple
- Mais ne fonctionne pas comme espéré
 - Trop lent
 - Pas fiable pour l'instant
- **En attendant une amélioration, j'interface le matériel avec Openhab**
- **Pas de problèmes avec Node-Red = recommandé**

Identification de la clef Xiaomi

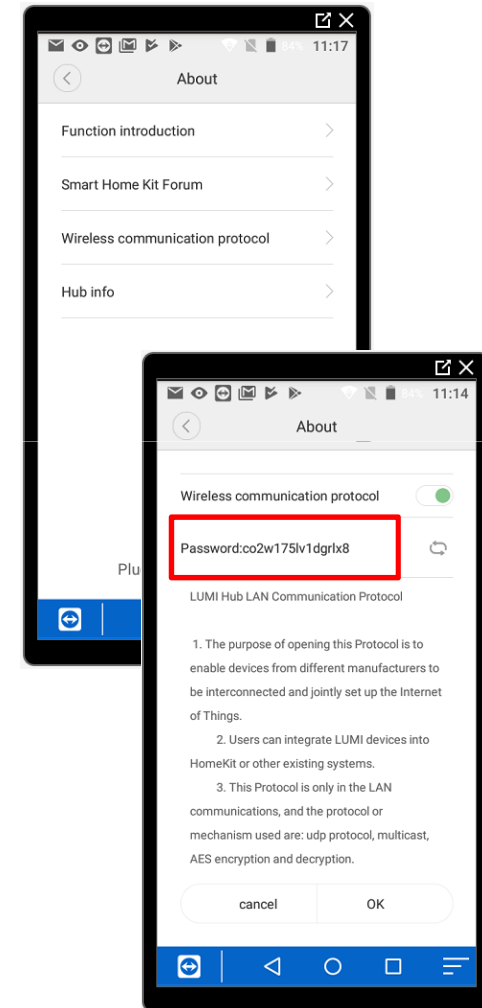
- Dans Gateway (EPFL)
- Cliquer sur « ... »
- Puis sur « About »



!!! Cette procédure est importante
si l'on veut renvoyer des données vers
le Gateway Mi-Home de Xiaomi

Identification de la clef Xiaomi

- Dans About sur
 - Wireless Communication Protocol
 - On y trouve le mot de passe du gateway
- On note le mot de passe pour la configuration dans Node-Red



FIN